Fiscal Year:	FY 2013	**Task Last Updated:**	FY 02/27/2013
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Project Title:	Sensitivity Modulation of Different T-Cells Activation Pathways by Microgravity (PADIAC)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology		
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Zip Code:	94121	**Congressional District:**	8
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Project Type:	Flight	**Solicitation / Funding Source:**	ESA-RA-LS-01-LSRA
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	Soyuz/ISS		
Flight Assignment:	ISS NOTE: New end date is 12/30/2013 per A. Chu/ARC and NSSC (Ed., 2/21/2013) NOTE: New end date is 6/30/2013 per A. Chu/ARC (Ed., 1/27/2012) NOTE: to be extended to 6/30/2013 per A. Chu/ARC (Ed., 1/17/2012) NOTE: Extended to 6/30/2012 (from 6/30/2010) per A. Chu/ARC (6/2010)		
Key Personnel Changes/Previous PI:	PI is Dr. Isabelle Walther of Switzerland. Dr. Hughes-Fulford is the US PI (ESA CoI).		
COI Name (Institution):	Walther, Isabelle (Swiss Federal Institute of Technology)		
Grant/Contract No.:	NNX07AM98G		
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Task Description:	This is an international collaboration between Dr. Millie Hughes-Fulford, Dr. Isabelle Walther, Augusto Cogoli, Proto Pippia, Mariantonia Meli, Grazia Galleri. Mammalian cells subjected to conditions of simulated microgravity on ground as well as to space flight conditions are showing alternations in their structure and function. Changes in proliferation, differentiation, and genetic expression have been demonstrated in several types of cells (reviewed in Cogoli 1996, Cogoli and Greuter 1997). Among the mammalian cells, the cells of the immune system, in particular, are severely affected by the space environment. Especially T-lymphocytes are altered in their activation process. Change in growth rate, cytokine production, locomotion, gene expression, and PKC distribution have been observed on Space Shuttle flights (Tipton et al. 1996, Pellis et al. 1997, Cogoli-Greuter et al. 1997, Hatton et al. 1999, Lewis et al. 2001). At the beginning of the space research, the experiments are based on simple question with answers like “yes/no”; at the present time the knowledge on the signal transduction mechanisms are progressing rapidly and the research is orienting to answer deeper questions about regulation pathways and to get a better understanding at the molecular level. Moreover, the laboratory techniques have evolved greatly. Sub-populations of T-cells can be isolated from peripheral blood, the expression of specific genes can be measured, and signal pathways can be specifically activated. This allows studying more complex and subtle reactions in cells biology. The goal of this proposal is to study, thanks to these new methods, the finer regulation of lymphocyte activation, especially at the level of the cytokines IL-2, IL-1, and their receptors using different activation pathways. The specific objectives are to further investigate, using several activators, the genetic expression of the co-stimulatory transduction pathway mediated by the CD28, to assess whether the IL-1 receptor is involved in the inhibition of the activation, to see if the activation is also restored in space as it is in simulated microgravity using anti-CD3/IL-2, and to determine which genes are then expressed under these special conditions. Microgravity will be used as an inhibitor of activation without having to use chemicals putatively damaging the cells. The main hypothesis to be tested is whether the restoring of the activation under the specific anti-CD3/IL-2 activation is due to a restoring of the receptor (IL-2R and/or IL-1R) gene expression. Correlated aspects such as the expression of early oncogenes and the production cytokines and their receptor will also be studies. The experiment will be based on an on-going ground-based research using the RPM as the main tool.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Immune response is a key element in continued health. It is known that there is a suppression of the the immune system during spaceflight. Our studies are aimed at understanding the molecular defect of the T cell that suppresses activation during microgravity. Once we understand the controlling pathways, by using microgravity as an investigative tool, we can identify pharmaceutical targets of the immune system. These targets will include those that enhance T cell function. as well as targets to blunt immune function to treat autoimmune diseases such as lupus and Rheumatoid arthritis.
	Sensitivity Modulation of Different T-cells Activation Pathways by Microgravity Millie Hughes-Fulford, Ph.D. Professor, UCSF Research Scientist, NCIRE VAMC 4150 Clement Street (Mail code 151F) San Francisco, CA 94121 415-750-6940 Fax: 415-750-6667 milliehf1@gmail.com Task Description: This experiment was designed to discover the changes in the T cell immune response in microgravity. T-cells are a type of white blood cells that help protect us from infections. Upon contact with the infectious agent they secrete a group of substances, called lymphokines that are necessary for an efficient immune response against viruses and bacteria. Interleukin-2 (IL-2) is one of the most important products of T-cells. It is possible to purify T-cells from the peripheral blood of a donor and study their function in vitro. Instead of a virus or a bacterium, an innocuous substance called mitogen may be used to simulate infection and activate the immune response. The PADIAC experiment is the continuation of the research on human immune system performed over the last 24 years by the Space Biology Group in Zurich. The expression of interleukin-2 receptor alpha (IL-2Ra) is an essential step in T-cells activation. Nearly a total loss of T-cell activation in real microgravity was discovered in a Spacelab flight in 1983 by Augusto Cogoli and was confirmed later by several spaceflight experiments. Without full activation, cells appear to undergo apoptosis. Inhibition of IL-2Ra expression is probably one of the causes of such loss of activation and cell death. The LEUKIN experiment flown in 2006 has shown that multiple genes that are upregulated in the 1g flight samples are not upregulated in the flight µg controls. We found that in early activation the differences in microgravity were largely due to binding and changes in signal transduction and regulation of transcription, what was expected since in the early activation steps activators are bonded and new genes are transcripted. The main objectives of this experiment were to compare, using several activators, the genetic expression of the genes involved in the co-stimulatory transduction pathway mediated by the CD28, to assess whether the IL-1 receptor is involved in the inhibition of the activation, to see if the activation is also restored in space as it is in simulated microgravity using anti-CD3/IL-2 and to determine which genes are than expressed under these special conditions. For this experiment PBLs were isolated from fresh human blood shortly before launch and T cells were purified by high affinity negative selection with Human T-cells Enrichment Columns (R&D Systems). The cells were suspended in medium at final cell concentration of 8.75x106 cells per ml, and were sealed in the PADIAC culture chambers. Quadruplicates were treated simultaneously. After a short pre-incubation at 37°C (1h30’), the T-cells were activated automatically (see activator below), and further incubated for 0h, 4, and 24h; finally the cells were fixed by addition of RNAlater solution in order to isolate RNA and determine genetic expression. 1. Anti-human CD3 coated to beads + mouse anti- human monoclonal antibody CD28. 2. Anti- human CD3 coated to beads + IL-2. 3. Mouse anti-human monoclonal antibody CD28 + Con A. The main hypothesis to be tested is whether the restoring of the activation under the specific anti-CD3/IL-2 activation is due to a restoring of the receptor (IL-2R and/or IL-1R) gene expression and which are the genes affected using the

different activator mixes. Correlated aspects such as the expression of early oncogenes and the production cytokines and their receptor will also be studied. We hypothesize that the gene expression of IL-2 and its IL-2a receptor unit is altered during spaceflight. With loss of activation, we hypothesize that the expression or activation of components of the signal transduction pathways will also be reduced in microgravity.

Research Objectives

The objectives of this experiment are: - To investigate the finer regulation of the T-cells activation using 3 different activator mixes ; - To assess whether the IL-1receptor is involved in the inhibition of the activation; - To determine if the expression of IL-2 receptor subunits is inhibited under microgravity ; - To identify which clusters of genes activated by the activator Concanavalin A / anti CD28 are sensitive to gravity ; -To see if the activation is restored in space as it is in simulated microgravity using anti CD3/IL2

To reach these objectives the RNA of the cells will be analysed by micro-array and real-time PCR.

Experiment Method / Protocol

The experiment was integrated into 13 KIC-SLE containers. The cells for the experiment are T-cells isolated from freshly drawn blood from healthy donors. The isolated cells were suspended into RPMI 1640/10% FCS. The cells were at a concentration of 8.75x10E6 per ml, giving 7x10E6 per wells. The 3 different activators were mixtures of ConcanavalinA/antiCD28; antiCD28/antiCD3 beads, and antiCD3bead/IL2. The fixative was RNAlater (product commercially available). The cells were isolated at L-2D, stored at RT and the units were filled at L-1D. The 2 flight pouches containing all units (7 + 6) packed inside warmpack element were handed over at L-14h and stored until transfer in ISS at RT. The warmpack elements kept the cells at a temperature between 23 and 37°C. After transfer to ISS the units were loaded into KUBIK-6 (in EDR) and KUBIK-3 (stand-alone) in static or centrifuged positions at 37°C. Once the experiment completed, the containers were stored at 4°C for 24h and finally transferred to MELFI until descent; from landing site, the containers were transported at cool temperature until Moscow and in frozen conditions until Zurich. Finally after opening the units in Zurich, the samples were sent frozen to our co-I in San Francisco.

The 6 units (3 static + 3 centrifuge) for the 24h incubation were located in KUB-6 (EDR). In the stand-alone, first the time 0h static unit was inserted, after 1h40 this one was removed and replaced by 3 static and 3 centrifuge units (4h incubation). The units were after fixation stored about 24h at 4°C then in MELFI at -80°C until descent at RT.

Analyses: Genetic expression of genes involved in the activation pathway.

List of Hardware Items used

For this experiment one hardware was used: The PAD hardware designed and constructed by Swiss companies. The hardware was placed inside the KIC-SLE container which was inserted for the incubation inside KUBIK 3 (stand-alone) and KUBIK 6 in EDR. For cool storage the TCU 2 from Biolab and the Melfi were used.

Relevance of project/Earth Benefit:

Signal transduction research is one of the most studied fields in biology today. Signal transduction cascades drive almost every cell's reaction. The results obtained in space with relatively simple experiments showed surprising reactions of the cells to this new environment. As space access is very limited and statistical significance is difficult, it is important to perform systematic study based on a well-anchored ground and flight research. In this project we intend to systematically investigate, at the molecular level, the T–cell activation focusing on the CD28 mediated pathway. The novelty of this proposal is that the cells will be activated by several activating cocktails, one of which can restore the activation in simulated microgravity. It is expected that new knowledge on the cause of the inhibition of the T-cells activation in space will be gathered thank the comparison, at the molecular level, between the different activation pathways. Moreover the up-to-now little investigated IL-1R will also be studied leading to a more defined understanding of its effect on the downstream signaling.

The main hypothesis to be tested is whether the restoring of the activation under the specific anti-CD3/IL-2 activation is due to a restoring of the receptor (IL-2R and/or IL-1R) gene expression. Earth Benefits of this type of research includes the ability to find new target genes for treatment of immune disease, since microgravity gives us a new window to understand the control mechanisms of T cell activation in a non-Earth condition.

Task Progress:

PROGRESS REPORT : The flight finally got off the ground in 2011 and data analysis showed differences between the different methods of T-cell activation (A1, A2, or A3). All three methods show some activation within the 4 hour timepoint. IL-2 is a limiting step since at 4 hours using IL-2 jumpstarts IL-2 gene expression.. By the time that IL2Ra is made (around 3 hours), all systems in gravity can make the rate limiting subunit; however, the microgravity samples are all reduced in response. Same is true for ford the other key regulators of the immune response XCL2, EGR1, CCL20, GMCSF, TNDa, IFNg.

Data from spaceflight using NT (non activated) or A1 treated (ConA/CD28, A2 using CD3/CD28 or A3 treated IL-2/CD3 show that IL-2 is a causal treatment for production of new IL-2 RNA, and that this is reduced with only treatments A1 or A2 are used, but is fully recovered when this regulator point is bypassed by adding IL-2 directly to the culture. When gravity is added, the gene expression of the TNFa, EGR1, CCL20, GMCSF and INFa is increased suggesting other regulatory molecules that are reduced by microgravity. Due to the long duration of sample delivery to ISS, it was not possible to use gene array analysis due to the quality of the RNA.

FUTURE PLANS: The final understanding will come from from protein analysis of the kinase pathways that are activated by the different methods. This is currently impossible in spaceflight since no hardware exists that allows collection of cells and preservation of proteins over a long period.. We will accomplish this using a ground based random positioning machine as a microgravity surrogate.

ABBREVIATIONS Con A, concanavalin A; ESA, European Space Agency; ETHZ, Swiss Federal Institute of Technology Zurich; IFN gamma, Interferon gamma; IL, Interleukin IL-1R, Interleukin-1 receptor; IL-2R, Interleukin-2 receptor; IML, International Microgravity Laboratory; MHC, major histocompatibility complex; NASA, National Aeronautics and Space Administration; PKA, protein kinase A; PKC, protein kinase C; RPM, random-positioning machine; RT-PCR, Reverse transcriptase polymerase chain reaction; TCR, T-cell receptor; TNF, tumor necrosis factor

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